

ISC Class XII Physics

Chapter 2 Electrostatic Potential, Potential Energy & Capacitance

69 Numericals Answer Key

Step-by-Step ISC Board Solutions

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| II. | Potential Gradient |
| III. | Electrostatic Potential Energy |
| IV. | Electric Dipole |
| V. | Capacitance of a Conductor |
| VI. | Parallel Plate Capacitor |
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Q.No.	Topic / Question Summary	Section
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Formula Sheet

I. Electric Potential and Potential Difference

Potential due to a point charge: $V = \frac{kQ}{r}$

$V \propto 1/r$; slope of a V vs $1/r$ graph gives kQ

Potential difference & work: $V = \frac{W}{q}$; $W = q(V_B - V_A)$

W = work done in moving charge q between two points

Potential due to several charges: $V = \sum_i \frac{kQ_i}{r_i}$

scalar sum (superposition), unlike vector sum for $\vec{E}$

Kinetic energy gained: $K = qV$

gain in KE = work done by the field for a charge accelerated through PD V

II. Potential Gradient

Potential gradient: $E = \frac{\Delta V}{d}$

electric field = (negative of) rate of change of potential with distance; potential decreases along $\vec{E}$

III. Electrostatic Potential Energy

PE of two point charges: $U = \frac{kq_1q_2}{r}$

energy needed to assemble the charge pair from infinity

Energy in electron-volt: $1 \text{ eV} = 1.6 \times 10^{-19} \text{ J}$

convenient energy unit at atomic scale

IV. Electric Dipole (Energy Aspects)

Dipole moment: $p = q \times 2l$

$2l$ = separation between the two charges of the dipole

Torque on a dipole: $\tau = pE \sin \theta$

maximum at $\theta = 90^\circ$: $\tau_{max} = pE$

PE of a dipole in a field: $U = -pE \cos \theta$

$U = 0$ at $\theta = 90^\circ$; minimum ($-pE$) at $\theta = 0^\circ$

Work done rotating a dipole: $W = pE(\cos \theta_1 - \cos \theta_2)$

work to rotate dipole from θ_1 to θ_2 in the field

V. Capacitance of a Conductor

Capacitance: $C = \frac{Q}{V}$

ratio of charge to potential; unit: farad (F)

Capacitance of an isolated sphere: $C = 4\pi\epsilon_0 R = \frac{R}{k}$

R = radius of the sphere

VI. Parallel Plate Capacitor

Parallel plate capacitance (air): $C = \frac{\epsilon_0 A}{d}$

A = plate area, d = plate separation

Charge on capacitor: $Q = CV$

V = potential difference applied across the plates

VII. Capacitor with Dielectric Slab

Capacitance with dielectric: $C = \frac{K\epsilon_0 A}{d}$

K = dielectric constant of the medium between the plates

Effect of dielectric on C , Q , V : Q = const if battery removed; V = const if battery connected

battery removed $\Rightarrow V$ falls, Q fixed; battery connected $\Rightarrow V$ fixed, Q rises

VIII. Energy of a Charged Capacitor

Energy stored: $U = \frac{1}{2}CV^2 = \frac{Q^2}{2C} = \frac{1}{2}QV$

use whichever form matches the known quantities (Q fixed or V fixed)

Energy density: $u = \frac{U}{\text{Volume}} = \frac{1}{2}\epsilon_0 E^2$

energy stored per unit volume of the field region

IX. Combination of Capacitors

Series combination: $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$

same charge Q on each capacitor; PDs add up

Parallel combination: $C_p = C_1 + C_2 + \dots$

same PD across each capacitor; charges add up

X. Miscellaneous

Potential of a charged sphere: $V_{surface} = \frac{kQ}{R}$; $V_{inside} = V_{surface}$ (constant)

for a hollow/solid conducting sphere, potential inside equals that on the surface

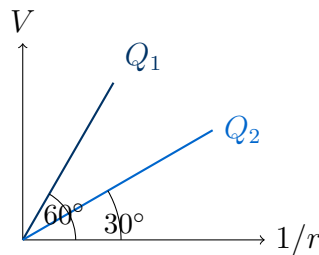
Balancing a charged drop: $qE = mg$

condition for a charged drop to remain stationary between plates

I. Electric Potential and Potential Difference

Question 1

The figure shows the variation of potential V with $\frac{1}{r}$ for two point charges Q_1 and Q_2 , where V is the potential at distance r . Find Q_1/Q_2 .



V vs 1/r graph: line for Q_1 at 60° , line for Q_2 at 30°

Given: Graph of V vs $\frac{1}{r}$: line for Q_1 has slope at 60° ; line for Q_2 has slope at 30°

Formula: $V = \frac{kQ}{r} = kQ \cdot \frac{1}{r}$, so slope of V - $\frac{1}{r}$ graph = kQ

Solution:

For a point charge, $V = kQ \cdot \frac{1}{r}$, so the slope of the V vs $\frac{1}{r}$ graph equals kQ .

$$\frac{Q_1}{Q_2} = \frac{\text{slope}_1}{\text{slope}_2} = \frac{\tan 60^\circ}{\tan 30^\circ} = \frac{\sqrt{3}}{1/\sqrt{3}} = 3$$

Final Answer

$$Q_1/Q_2 = 3 : 1$$

Question 2

The amount of work done in bringing a charge of 1.3×10^{-7} C from infinity to a point in an electric field is 6.5×10^{-5} J. Find the electric potential at the point.

Given: $W = 6.5 \times 10^{-5}$ J; $q = 1.3 \times 10^{-7}$ C

Formula: $V = \frac{W}{q}$

Solution:

$$V = \frac{W}{q} = \frac{6.5 \times 10^{-5}}{1.3 \times 10^{-7}} = \frac{6.5}{1.3} \times 10^2 = 5 \times 10^2 = 500 \text{ V}$$

Final Answer

$$V = 500 \text{ V}$$

Question 3

5 J work is done in moving a positive charge of 0.5 C between two points. Find the potential difference between the points.

Given: $W = 5 \text{ J}$; $q = 0.5 \text{ C}$

Formula: $V = W/q$

Solution:

$$V = \frac{W}{q} = \frac{5}{0.5} = 10 \text{ V}$$

Final Answer

$$V_{AB} = 10 \text{ V}$$

Question 4

A wire is bent in a circle of radius 10 cm. It is given a charge of $250 \mu\text{C}$ which spreads uniformly. What is the electric potential and intensity of field at its centre?

Given: $r = 10 \text{ cm} = 0.10 \text{ m}$; $Q = 250 \times 10^{-6} \text{ C}$; $k = 9 \times 10^9 \text{ N m}^2\text{C}^{-2}$

Formula: $V = \frac{kQ}{r}$; $E = 0$ at centre of a uniformly charged ring (by symmetry)

Solution:

$$V = \frac{kQ}{r} = \frac{9 \times 10^9 \times 250 \times 10^{-6}}{0.10} = \frac{9 \times 10^9 \times 2.5 \times 10^{-4}}{0.10} = 2.25 \times 10^7 \text{ V}$$

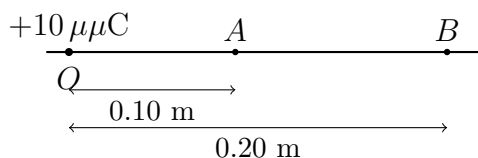
For a uniformly charged ring, every element's field contribution at the centre cancels its diametrically opposite element. Therefore $E_{\text{centre}} = 0$.

Final Answer

$$V = 2.25 \times 10^7 \text{ V}; \quad E = 0$$

Question 5

Calculate (i) electric potentials at points A and B , (ii) work done in taking the same amount of negative charge from A to B , given $OA = 0.10 \text{ m}$, $OB = 0.20 \text{ m}$ and a charge of $+10 \mu\text{C}$ at O .



Charge $+10 \mu\text{C}$ at O , with points A and B along a line

Given: $Q = +10 \text{ pC} = +10 \times 10^{-12} \text{ C}$; $OA = 0.10 \text{ m}$; $OB = 0.20 \text{ m}$; $k = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$

Formula: $V = \frac{kQ}{r}$; $W = q(V_B - V_A)$, where q is the charge being moved

Solution:

(i) Potentials:

$$V_A = \frac{kQ}{OA} = \frac{9 \times 10^9 \times 10^{-11}}{0.10} = \frac{0.09}{0.10} = 0.90 \text{ V}$$

$$V_B = \frac{kQ}{OB} = \frac{9 \times 10^9 \times 10^{-11}}{0.20} = \frac{0.09}{0.20} = 0.45 \text{ V}$$

(ii) Work done on the negative charge ($q = -10^{-11} \text{ C}$) from A to B :

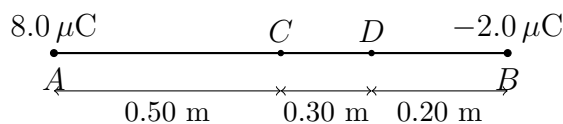
$$W = q(V_B - V_A) = (-10^{-11})(0.45 - 0.90) = (-10^{-11})(-0.45) = 4.5 \times 10^{-12} \text{ J}$$

Final Answer

(i) $V_A = 0.90 \text{ V}$, $V_B = 0.45 \text{ V}$; (ii) $W = 4.5 \times 10^{-12} \text{ J}$

Question 6

A charge A of $8.0 \mu\text{C}$ is situated 1.00 m from a charge B of $-2.0 \mu\text{C}$. Calculate the potential at the midpoint C between A and B . Also find the potential at D (0.80 m from A , 0.20 m from B). How much work will be done in taking $0.02 \mu\text{C}$ from D to C ?



Charges $8.0 \mu\text{C}$ at A and $-2.0 \mu\text{C}$ at B , with points C (midpoint) and D on the line AB

Given: $Q_A = +8.0 \times 10^{-6} \text{ C}$; $Q_B = -2.0 \times 10^{-6} \text{ C}$; $AB = 1.00 \text{ m}$; $AC = BC = 0.50 \text{ m}$; $AD = 0.80 \text{ m}$, $BD = 0.20 \text{ m}$; $q = 0.02 \times 10^{-6} \text{ C}$; $k = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$

Formula: $V = \sum \frac{kQ_i}{r_i}$; $W = q(V_C - V_D)$

Solution:**Potential at C (midpoint, $r = 0.50$ m from each):**

$$V_C = k \left(\frac{Q_A}{0.50} + \frac{Q_B}{0.50} \right) = 9 \times 10^9 \left(\frac{8 \times 10^{-6}}{0.50} + \frac{-2 \times 10^{-6}}{0.50} \right) = 9 \times 10^9 \times \frac{6 \times 10^{-6}}{0.50} = 1.08 \times 10^5 \text{ V}$$

Potential at D ($AD = 0.80$ m, $BD = 0.20$ m):

$$V_D = k \left(\frac{Q_A}{0.80} + \frac{Q_B}{0.20} \right) = 9 \times 10^9 \left(\frac{8 \times 10^{-6}}{0.80} + \frac{-2 \times 10^{-6}}{0.20} \right) = 9 \times 10^9 (10^{-5} - 10^{-5}) = 0 \text{ V}$$

Work done taking $q = 0.02 \mu\text{C}$ from D to C:

$$W = q(V_C - V_D) = 0.02 \times 10^{-6} \times (1.08 \times 10^5 - 0) = 2.16 \times 10^{-3} \text{ J}$$

Final Answer

$$V_C = 1.08 \times 10^5 \text{ V}; \quad V_D = 0 \text{ V}; \quad W = 2.16 \times 10^{-3} \text{ J}$$

★ Likely Exam Question**Question 7**

Two charges $+20 \mu\text{C}$ and $-10 \mu\text{C}$ are 6 cm apart. Find the electric potential at a point distant 4 cm on the right bisector of the line joining the charges.

Given: $Q_1 = +20 \times 10^{-6} \text{ C}$; $Q_2 = -10 \times 10^{-6} \text{ C}$; separation = 6 cm = 0.06 m; perpendicular distance = 4 cm = 0.04 m; $k = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$

Formula: $V = \sum \frac{kQ_i}{r_i}$

Solution:

Distance from each charge to the point P on the bisector (half-separation = 3 cm):

$$r = \sqrt{(3)^2 + (4)^2} \text{ cm} = \sqrt{9 + 16} \text{ cm} = 5 \text{ cm} = 0.05 \text{ m}$$

$$V = k \left(\frac{Q_1}{r} + \frac{Q_2}{r} \right) = \frac{k}{r} (Q_1 + Q_2) = \frac{9 \times 10^9}{0.05} (20 - 10) \times 10^{-6} = \frac{9 \times 10^9 \times 10^{-5}}{0.05} = 1.8 \times 10^6 \text{ V}$$

Final Answer

$$V = 1.8 \times 10^6 \text{ V}$$

Question 8

A point charge of $6.0 \times 10^{-8} \text{ C}$ is situated at the coordinate origin. How much work will be done in taking an electron from the point $x = 3 \text{ m}$ to $x = 6 \text{ m}$?

Given: $Q = 6.0 \times 10^{-8} \text{ C}$; $r_1 = 3 \text{ m}$; $r_2 = 6 \text{ m}$; $q_e = -1.6 \times 10^{-19} \text{ C}$; $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

Formula: $W = q_e(V_2 - V_1)$; $V = kQ/r$

Solution:

$$V_1 = \frac{kQ}{r_1} = \frac{9 \times 10^9 \times 6 \times 10^{-8}}{3} = \frac{540}{3} = 180 \text{ V}$$

$$V_2 = \frac{kQ}{r_2} = \frac{9 \times 10^9 \times 6 \times 10^{-8}}{6} = \frac{540}{6} = 90 \text{ V}$$

$$W = q_e(V_2 - V_1) = (-1.6 \times 10^{-19})(90 - 180) = (-1.6 \times 10^{-19})(-90) = 1.44 \times 10^{-17} \text{ J}$$

Final Answer

$$W = 1.44 \times 10^{-17} \text{ J}$$

Question 9

At each corner of an equilateral triangle, charges of $+250 \mu\text{C}$ are placed in air. What will be the resultant electric potential at the centroid of the triangle whose distance from each corner is 18 cm ?

Given: $Q = 250 \times 10^{-6} \text{ C}$ at each corner; $r = 18 \text{ cm} = 0.18 \text{ m}$; $k = 9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$

Formula: $V = \sum \frac{kQ_i}{r_i}$; since all charges are equal and equidistant, $V = 3 \times \frac{kQ}{r}$

Solution:

All three charges are equal and equidistant from the centroid:

$$V = 3 \times \frac{kQ}{r} = 3 \times \frac{9 \times 10^9 \times 250 \times 10^{-6}}{0.18} = 3 \times \frac{2.25 \times 10^6}{0.18} = 3 \times 1.25 \times 10^7 = 3.75 \times 10^7 \text{ V}$$

Final Answer

$$V = 3.75 \times 10^7 \text{ V}$$

Question 10 (ISC 2008)

How much kinetic energy is gained by an electron, initially at rest, when it moves through a potential difference of 1000 V ? Write your answer in joule.

Given: $q_e = 1.6 \times 10^{-19} \text{ C}$; $V = 1000 \text{ V}$

Formula: $K = qV$

Solution:

$$K = qV = (1.6 \times 10^{-19}) \times 1000 = 1.6 \times 10^{-16} \text{ J}$$

Final Answer

$$K = (1.6 \times 10^{-19}) \times 1000 \text{ J} = 1.6 \times 10^{-16} \text{ J}$$

★ Likely Exam Question**Question 11**

An α -particle at rest is accelerated through 10^6 V . (i) What will be its kinetic energy? (ii) If a proton is accelerated through the same potential difference, how much kinetic energy will it gain? (iii) Which particle will gain more velocity if both start from rest?

Given: $V = 10^6 \text{ V}$; $q_\alpha = 2e = 3.2 \times 10^{-19} \text{ C}$; $q_p = e = 1.6 \times 10^{-19} \text{ C}$; $m_\alpha = 6.64 \times 10^{-27} \text{ kg}$; $m_p = 1.67 \times 10^{-27} \text{ kg}$

Formula: $K = qV$; $v = \sqrt{2K/m}$

Solution:

(i) **KE of α -particle:**

$$K_\alpha = q_\alpha V = 2 \times 1.6 \times 10^{-19} \times 10^6 = 3.2 \times 10^{-13} \text{ J}$$

(ii) **KE of proton:**

$$K_p = q_p V = 1.6 \times 10^{-19} \times 10^6 = 1.6 \times 10^{-13} \text{ J}$$

(iii) **Velocity comparison:**

Since $K = \frac{1}{2}mv^2$, we get $v = \sqrt{2K/m}$.

$$v_\alpha = \sqrt{\frac{2 \times 3.2 \times 10^{-13}}{6.64 \times 10^{-27}}} \approx 9.8 \times 10^6 \text{ m/s}$$

$$v_p = \sqrt{\frac{2 \times 1.6 \times 10^{-13}}{1.67 \times 10^{-27}}} \approx 1.38 \times 10^7 \text{ m/s}$$

The **proton** gains more velocity because, despite having smaller KE, it has a much smaller mass.

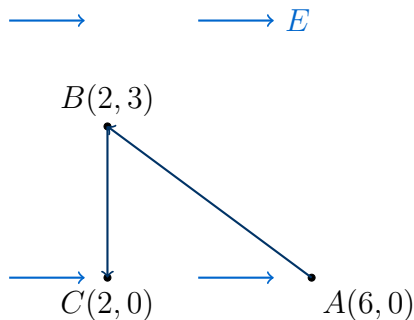
Final Answer

(i) $K_\alpha = 3.2 \times 10^{-13} \text{ J}$; (ii) $K_p = 1.6 \times 10^{-13} \text{ J}$; (iii) Proton gains more velocity

Question 12

A test charge q is moved without acceleration from A to C along the path $A \rightarrow B \rightarrow C$ in electric field E as shown in the figure. (i) Calculate the potential difference between

A and C . (ii) At which point (A or C) is the electric potential more and why?



Path $A(6, 0) \rightarrow B(2, 3) \rightarrow C(2, 0)$ in a uniform field $\vec{E}$ along $+x$

Given: $A = (6, 0)$; $B = (2, 3)$; $C = (2, 0)$; $\vec{E}$ points in the $+x$ direction

Formula: $V_A - V_C = E \cdot d_x$, where d_x is displacement along $\vec{E}$; potential decreases in direction of $\vec{E}$

Solution:

(i) Only the component of displacement parallel to $\vec{E}$ (the x -direction) contributes to potential difference. The x -coordinates: $x_A = 6$, $x_C = 2$, so the charge moved a distance $\Delta x = 4$ units against $\vec{E}$ (from C to A).

$$V_C - V_A = E \cdot (x_A - x_C) = 4E$$

(ii) Since C is at $x = 2 < x_A = 6$, and potential increases opposite to $\vec{E}$, we have $V_C > V_A$.

Final Answer

(i) $V_C - V_A = 4E$; (ii) $V_C > V_A$, because potential decreases in the direction of $\vec{E}$

II. Potential Gradient

Question 13

10 J of work is required to move a charge of -200 C from point A to B . (i) Which of the two points is at higher potential? (ii) What is the PD between them? (iii) If the points are 4 cm apart, what is the component of the electric field parallel to the line joining A and B ?

Given: $W = 10\text{ J}$; $q = -200\text{ C}$; $d = 4\text{ cm} = 0.04\text{ m}$

Formula: $W = q(V_B - V_A)$; $E = \frac{\Delta V}{d}$

Solution:

(i) Using $W = q(V_B - V_A)$:

$$V_B - V_A = \frac{W}{q} = \frac{10}{-200} = -0.05\text{ V}$$

Since $V_B - V_A < 0$, we have $V_A > V_B$. Point **A** is at higher potential.

(ii) $|V_A - V_B| = 0.05\text{ V}$

(iii)

$$E = \frac{|V_A - V_B|}{d} = \frac{0.05}{0.04} = 1.25\text{ V m}^{-1}$$

Final Answer

(i) A is at higher potential; (ii) $PD = 0.05\text{ V}$; (iii) $E = 1.25\text{ V m}^{-1}$

III. Electrostatic Potential Energy

Question 14

The mutual electric potential energy between two protons in a uranium nucleus is 2.4×10^5 eV. Find the distance between the protons.

Given: $U = 2.4 \times 10^5$ eV $= 2.4 \times 10^5 \times 1.6 \times 10^{-19}$ J $= 3.84 \times 10^{-14}$ J; $q_p = 1.6 \times 10^{-19}$ C; $k = 9 \times 10^9$ N m²C⁻²

Formula: $U = \frac{kq_1q_2}{r}$

Solution:

$$r = \frac{kq_p^2}{U} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{3.84 \times 10^{-14}} = \frac{9 \times 10^9 \times 2.56 \times 10^{-38}}{3.84 \times 10^{-14}} = \frac{2.304 \times 10^{-28}}{3.84 \times 10^{-14}} = 6.0 \times 10^{-15} \text{ m}$$

Final Answer

$$r = 6.0 \times 10^{-15} \text{ m}$$

Question 15

An electron is circulating around the nucleus of a hydrogen atom in a circular orbit of radius 5.3×10^{-11} m. Calculate the electric potential at this radius and electric potential energy of the atom in eV. What would be the electric potential due to a helium nucleus at the same radius?

Given: $r = 5.3 \times 10^{-11}$ m; $q_p = e = 1.6 \times 10^{-19}$ C; $q_{He} = 2e = 3.2 \times 10^{-19}$ C; $k = 9 \times 10^9$ N m²C⁻²

Formula: $V = \frac{ke}{r}$; $U = (-e) \times V$; $1 \text{ eV} = 1.6 \times 10^{-19}$ J

Solution:

Electric potential due to proton (H nucleus):

$$V_H = \frac{ke}{r} = \frac{9 \times 10^9 \times 1.6 \times 10^{-19}}{5.3 \times 10^{-11}} = \frac{1.44 \times 10^{-9}}{5.3 \times 10^{-11}} = 27.2 \text{ V}$$

Potential energy of hydrogen atom (electron in field of proton):

$$U = (-e) \times V_H = -1.6 \times 10^{-19} \times 27.2 = -4.352 \times 10^{-18} \text{ J}$$

$$U = \frac{-4.352 \times 10^{-18}}{1.6 \times 10^{-19}} = -27.2 \text{ eV}$$

Potential due to helium nucleus ($Z = 2$):

$$V_{He} = \frac{k(2e)}{r} = 2 \times V_H = 2 \times 27.2 = 54.4 \text{ V}$$

Final Answer

$$V_H = 27.2 \text{ V}; \quad U = -27.2 \text{ eV}; \quad V_{He} = 54.4 \text{ V}$$

Question 16

The distance between two protons is $1.0 \times 10^{-10} \text{ m}$. If they are made free, what will be the kinetic energy of each when they are very far apart? If one proton is kept fixed and the other is freed, what will be its kinetic energy?

Given: $r = 1.0 \times 10^{-10} \text{ m}; \quad q_p = 1.6 \times 10^{-19} \text{ C}; \quad k = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$

Formula: $U = \frac{kq^2}{r}$; conservation of energy: initial PE converts to KE

Solution:

Initial mutual potential energy:

$$U = \frac{kq_p^2}{r} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{1.0 \times 10^{-10}} = \frac{9 \times 10^9 \times 2.56 \times 10^{-38}}{10^{-10}} = 2.304 \times 10^{-18} \text{ J}$$

Case 1 (both free): By symmetry, each proton gains equal KE:

$$K_{\text{each}} = \frac{U}{2} = \frac{2.304 \times 10^{-18}}{2} = 1.15 \times 10^{-18} \text{ J}$$

Case 2 (one fixed, one free): All PE converts to KE of the free proton:

$$K = U = 2.30 \times 10^{-18} \text{ J}$$

Final Answer

Case 1: $K_{\text{each}} = 1.15 \times 10^{-18} \text{ J}$; Case 2: $K = 2.30 \times 10^{-18} \text{ J}$

Question 17

Two electrons are released towards each other with equal velocities of 10^6 m s^{-1} . What will be the closest approach between them?

Given: $v = 10^6 \text{ m s}^{-1}; \quad m_e = 9.1 \times 10^{-31} \text{ kg}; \quad e = 1.6 \times 10^{-19} \text{ C}; \quad k = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$

Formula: At closest approach both electrons stop momentarily: Total initial KE = PE at closest approach

Solution:

$$2 \times \frac{1}{2} m_e v^2 = \frac{ke^2}{r_{\min}}$$

$$\begin{aligned} r_{\min} &= \frac{ke^2}{m_e v^2} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})^2}{9.1 \times 10^{-31} \times (10^6)^2} \\ &= \frac{9 \times 10^9 \times 2.56 \times 10^{-38}}{9.1 \times 10^{-31} \times 10^{12}} = \frac{2.304 \times 10^{-28}}{9.1 \times 10^{-19}} = 2.53 \times 10^{-10} \text{ m} \approx 2.56 \times 10^{-10} \text{ m} \end{aligned}$$

Final Answer

$$r_{\min} \approx 2.56 \times 10^{-10} \text{ m}$$

IV. Electric Dipole

Question 18

Two point charges of $+1.0 \times 10^{-6}$ C and -1.0×10^{-6} C are 2.0 cm apart, forming an electric dipole in a uniform electric field of 1.0×10^5 V m $^{-1}$. What will be the maximum torque acting on it due to the field? What will be the work done for rotating it through 180° from the equilibrium position?

Given: $q = 1.0 \times 10^{-6}$ C; $2l = 2.0$ cm = 0.02 m; $E = 1.0 \times 10^5$ V m $^{-1}$

Formula: $p = q(2l)$; $\tau_{\max} = pE$ (at $\theta = 90^\circ$); $W = pE(\cos \theta_1 - \cos \theta_2)$

Solution:

$$p = q \times 2l = 1.0 \times 10^{-6} \times 0.02 = 2.0 \times 10^{-8} \text{ C m}$$

$$\tau_{\max} = pE = 2.0 \times 10^{-8} \times 1.0 \times 10^5 = 2.0 \times 10^{-3} \text{ N m}$$

Work done rotating from $\theta_1 = 0^\circ$ (stable equilibrium) to $\theta_2 = 180^\circ$:

$$W = pE(\cos \theta_1 - \cos \theta_2) = pE(\cos 0^\circ - \cos 180^\circ) = pE(1 - (-1)) = 2pE$$

$$W = 2 \times 2.0 \times 10^{-3} = 4.0 \times 10^{-3} \text{ J}$$

Final Answer

$$\tau_{\max} = 2.0 \times 10^{-3} \text{ N m}; \quad W = 4.0 \times 10^{-3} \text{ J}$$

★ Likely Exam Question

Question 19

An electric dipole of length 2 cm is placed with its axis making an angle of 60° to a uniform electric field of 10^5 N C $^{-1}$. If it experiences a torque of $8\sqrt{3}$ N m, calculate (i) magnitude of the charge on the dipole and (ii) potential energy of the dipole.

Given: $2l = 2$ cm = 0.02 m; $\theta = 60^\circ$; $E = 10^5$ N C $^{-1}$; $\tau = 8\sqrt{3}$ N m

Formula: $\tau = pE \sin \theta$; $U = -pE \cos \theta$

Solution:

(i) Dipole moment:

$$p = \frac{\tau}{E \sin \theta} = \frac{8\sqrt{3}}{10^5 \times \sin 60^\circ} = \frac{8\sqrt{3}}{10^5 \times \frac{\sqrt{3}}{2}} = \frac{8\sqrt{3} \times 2}{10^5 \times \sqrt{3}} = \frac{16}{10^5} = 1.6 \times 10^{-4} \text{ C m}$$

$$q = \frac{p}{2l} = \frac{1.6 \times 10^{-4}}{0.02} = 8 \times 10^{-3} \text{ C}$$

(ii) Potential energy:

$$U = -pE \cos \theta = -1.6 \times 10^{-4} \times 10^5 \times \cos 60^\circ = -1.6 \times 10^{-4} \times 10^5 \times 0.5 = -8 \text{ J}$$

Final Answer

(i) $q = 8 \times 10^{-3} \text{ C}$; (ii) $U = -8 \text{ J}$

V. Capacitance of a Conductor

Question 20

A conductor when charged to 1000 V takes $2 \mu\text{C}$ of charge. Find the capacity of the conductor.

Given: $V = 1000 \text{ V}$; $Q = 2 \times 10^{-6} \text{ C}$

Formula: $C = Q/V$

Solution:

$$C = \frac{Q}{V} = \frac{2 \times 10^{-6}}{1000} = 2 \times 10^{-9} \text{ F} = 2 \times 10^{-3} \mu\text{F}$$

Final Answer

$$C = 2 \times 10^{-9} \text{ F} = 2 \times 10^{-3} \mu\text{F}$$

Question 21

The capacitance of a spherical conductor is $1 \mu\text{F}$. What is its diameter?

Given: $C = 1 \times 10^{-6} \text{ F}$; $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $C = 4\pi\epsilon_0 R = R/k$ (since $k = 1/4\pi\epsilon_0$)

Solution:

$$R = kC = 9 \times 10^9 \times 1 \times 10^{-6} = 9000 \text{ m}$$

$$\text{Diameter} = 2R = 18000 \text{ m} = 1.8 \times 10^4 \text{ m}$$

Final Answer

$$\text{Diameter} = 1.8 \times 10^4 \text{ m}$$

Common Mistake: A sphere of capacitance $1 \mu\text{F}$ would have a diameter of about 18 km — nearly the size of a small planet! Real conductors have capacitances in the pF range.

VI. Parallel Plate Capacitor

Question 22

A PD of 250 V exists across the plates of a $25 \mu\text{F}$ capacitor. Find the charge on each plate.

Given: $V = 250 \text{ V}$; $C = 25 \times 10^{-6} \text{ F}$

Formula: $Q = CV$

Solution:

$$Q = CV = 25 \times 10^{-6} \times 250 = 6.25 \times 10^{-3} \text{ C}$$

Plates carry $+6.25 \times 10^{-3} \text{ C}$ and $-6.25 \times 10^{-3} \text{ C}$.

Final Answer

$Q = +6.25 \times 10^{-3} \text{ C}$ and $Q = -6.25 \times 10^{-3} \text{ C}$

Question 23

What distance d between the two parallel plates, each of area $A = 0.0314 \text{ m}^2$, of an air capacitor can be kept so that its capacitance is the same as that of a conducting sphere of radius $a = 0.5 \text{ m}$?

Given: $A = 0.0314 \text{ m}^2$; $a = 0.5 \text{ m}$; $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $C_{\text{sphere}} = 4\pi\epsilon_0 a$; $C_{\text{plate}} = \epsilon_0 A/d$; set equal: $d = A/(4\pi a)$

Solution:

Setting capacitances equal:

$$\frac{\epsilon_0 A}{d} = 4\pi\epsilon_0 a \implies d = \frac{A}{4\pi a}$$

$$d = \frac{0.0314}{4 \times 3.14159 \times 0.5} = \frac{0.0314}{6.2832} = 5.0 \times 10^{-3} \text{ m} = 5.0 \text{ mm}$$

Final Answer

$d = 5.0 \text{ mm}$

Question 24

An air parallel-plate capacitor has a plate area of $6 \times 10^{-3} \text{ m}^2$ and a plate separation of 3 mm. Calculate its capacitance. If this capacitor is connected to a 100 V supply, what is the charge on each plate of the capacitor?

Given: $A = 6 \times 10^{-3} \text{ m}^2$; $d = 3 \text{ mm} = 3 \times 10^{-3} \text{ m}$; $V = 100 \text{ V}$; $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $C = \epsilon_0 A/d$; $Q = CV$

Solution:

$$C = \frac{\epsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 6 \times 10^{-3}}{3 \times 10^{-3}} = \frac{8.85 \times 10^{-12} \times 6}{3} = 17.7 \times 10^{-12} \text{ F} \approx 18 \text{ pF}$$

$$Q = CV = 18 \times 10^{-12} \times 100 = 1.8 \times 10^{-9} \text{ C}$$

Final Answer

$C \approx 18 \text{ pF}$; $Q = 1.8 \times 10^{-9} \text{ C}$

VII. Parallel Plate Capacitor with Dielectric Slab between Plates

Question 25

The area of each plate of a parallel-plate capacitor is 100 cm^2 and the distance between them is 0.05 cm . When it is filled with a dielectric, its capacitance becomes $3.54 \times 10^{-10} \text{ F}$. Find the dielectric constant.

Given: $A = 100 \text{ cm}^2 = 10^{-2} \text{ m}^2$; $d = 0.05 \text{ cm} = 5 \times 10^{-4} \text{ m}$; $C = 3.54 \times 10^{-10} \text{ F}$;
 $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $C = K\epsilon_0 A/d$; $K = Cd/(\epsilon_0 A)$

Solution:

$$K = \frac{Cd}{\epsilon_0 A} = \frac{3.54 \times 10^{-10} \times 5 \times 10^{-4}}{8.85 \times 10^{-12} \times 10^{-2}} = \frac{1.77 \times 10^{-13}}{8.85 \times 10^{-14}} = 2.0$$

Final Answer

Dielectric constant $K = 2.0$

Question 26

How much area of paper will be required to construct a parallel-plate capacitor of $0.004 \mu\text{F}$, if the dielectric constant of paper is 2.5 and its thickness is 0.025 mm ?

Given: $C = 4 \times 10^{-9} \text{ F}$; $K = 2.5$; $d = 0.025 \text{ mm} = 2.5 \times 10^{-5} \text{ m}$; $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $C = K\epsilon_0 A/d$; $A = Cd/(K\epsilon_0)$

Solution:

$$A = \frac{Cd}{K\epsilon_0} = \frac{4 \times 10^{-9} \times 2.5 \times 10^{-5}}{2.5 \times 8.85 \times 10^{-12}} = \frac{10^{-13}}{22.125 \times 10^{-12}} \approx 4.52 \times 10^{-3} \text{ m}^2$$

Final Answer

$A \approx 4.52 \times 10^{-3} \text{ m}^2$

Question 27

When a 20 V battery is connected to an air-capacitor, a charge of $30 \mu\text{C}$ is obtained. When oil is filled between the plates of the capacitor, a charge of $75 \mu\text{C}$ is obtained. Calculate (i) the dielectric constant of the oil, (ii) the energy contained in the oil-capacitor.

Given: $V = 20 \text{ V}$; $Q_{\text{air}} = 30 \times 10^{-6} \text{ C}$; $Q_{\text{oil}} = 75 \times 10^{-6} \text{ C}$

Formula: $K = C_{\text{oil}}/C_{\text{air}} = Q_{\text{oil}}/Q_{\text{air}}$ (same V); $U = \frac{1}{2}C_{\text{oil}}V^2$

Solution:

(i)

$$K = \frac{Q_{\text{oil}}}{Q_{\text{air}}} = \frac{75}{30} = 2.5$$

(ii)

$$C_{\text{oil}} = \frac{Q_{\text{oil}}}{V} = \frac{75 \times 10^{-6}}{20} = 3.75 \times 10^{-6} \text{ F}$$

$$U = \frac{1}{2}C_{\text{oil}}V^2 = \frac{1}{2} \times 3.75 \times 10^{-6} \times (20)^2 = \frac{1}{2} \times 3.75 \times 10^{-6} \times 400 = 7.5 \times 10^{-4} \text{ J}$$

Final Answer

(i) $K = 2.5$; (ii) $U = 7.5 \times 10^{-4} \text{ J}$

Question 28

We have a glass slab ($K = 7$) 4.00 mm thick, a mica foil ($K = 6$) 0.20 mm thick and an amber plate ($K = 2$) 2.0 cm thick. Which one should be placed between the plates of a parallel-plate capacitor to obtain maximum capacitance?

Formula: $C = K\varepsilon_0 A/d$, so maximising C for a given plate area means maximising K/d .

Solution:

Compare K/d for each material (all thicknesses in mm):

$$\text{Glass: } K/d = 7/4.00 \text{ mm}^{-1} = 1.75 \text{ mm}^{-1}$$

$$\text{Mica: } K/d = 6/0.20 \text{ mm}^{-1} = 30.0 \text{ mm}^{-1}$$

$$\text{Amber: } K/d = 2/20.0 \text{ mm}^{-1} = 0.10 \text{ mm}^{-1}$$

Mica has the largest K/d , so it gives the maximum capacitance.

Final Answer

Mica foil should be placed between the plates for maximum capacitance.

Question 29

The potential difference between the plates of a parallel-plate capacitor is 200 V. Area of each plate is 100 cm^2 and the distance between them is 1 mm. If the medium between the plates is air, calculate the charge taken by the capacitor. If there is a medium of dielectric constant 2.5 between the plates, what will be the potential difference for the same charge?

Given: $V_{\text{air}} = 200 \text{ V}$; $A = 100 \text{ cm}^2 = 10^{-2} \text{ m}^2$; $d = 1 \text{ mm} = 10^{-3} \text{ m}$; $K = 2.5$;
 $\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $C = \varepsilon_0 A/d$; $Q = CV$; $C' = KC$; $V' = Q/C'$

Solution:

$$C_{\text{air}} = \frac{\varepsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 10^{-2}}{10^{-3}} = 8.85 \times 10^{-11} \text{ F}$$

$$Q = C_{\text{air}} \times V = 8.85 \times 10^{-11} \times 200 = 1.77 \times 10^{-8} \text{ C}$$

With dielectric (Q unchanged, charge conserved):

$$C' = KC_{\text{air}} = 2.5 \times 8.85 \times 10^{-11} = 2.2125 \times 10^{-10} \text{ F}$$

$$V' = \frac{Q}{C'} = \frac{1.77 \times 10^{-8}}{2.2125 \times 10^{-10}} = 80 \text{ V}$$

Final Answer

$Q = 1.77 \times 10^{-8} \text{ C}$; new $V' = 80 \text{ V}$

Question 30

A paper is placed between two plates of copper, area of each 100 cm^2 . The thickness of the paper is 0.005 cm and its dielectric constant is 2.5 . If the paper can bear an electric field of $5 \times 10^5 \text{ V/cm}$, then calculate the maximum voltage to which the capacitor can be charged. How much charge will be stored on the capacitor?

Given: $A = 10^{-2} \text{ m}^2$; $d = 0.005 \text{ cm} = 5 \times 10^{-5} \text{ m}$; $K = 2.5$; $E_{\text{max}} = 5 \times 10^5 \text{ V/cm} = 5 \times 10^7 \text{ V/m}$; $\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $V_{\text{max}} = E_{\text{max}} \times d$; $C = K\varepsilon_0 A/d$; $Q = CV_{\text{max}}$

Solution:

$$V_{\text{max}} = E_{\text{max}} \times d = 5 \times 10^7 \times 5 \times 10^{-5} = 2500 \text{ V}$$

$$C = \frac{K\varepsilon_0 A}{d} = \frac{2.5 \times 8.85 \times 10^{-12} \times 10^{-2}}{5 \times 10^{-5}} = \frac{2.2125 \times 10^{-13}}{5 \times 10^{-5}} = 4.425 \times 10^{-9} \text{ F}$$

$$Q = CV_{\text{max}} = 4.425 \times 10^{-9} \times 2500 = 1.106 \times 10^{-5} \text{ C} \approx 1.1 \times 10^{-5} \text{ C}$$

Final Answer

$V_{\text{max}} = 2500 \text{ V}$; $Q \approx 1.1 \times 10^{-5} \text{ C}$

Question 31

A parallel-plate air capacitor has a plate area of $6 \times 10^{-3} \text{ m}^2$ and a plate separation of 3 mm. A PD of 100 V is established by a battery. After disconnecting the battery, the space between the plates is filled by a 3 mm thick mica sheet ($K = 6$). Find (i) the new PD, (ii) the final capacitance and (iii) the final charge. Also answer if battery remains connected.

Given: $A = 6 \times 10^{-3} \text{ m}^2$; $d = 3 \times 10^{-3} \text{ m}$; $V_0 = 100 \text{ V}$; $K = 6$; $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $C_0 = \epsilon_0 A/d$; $C' = KC_0$; Disconnected: $Q = \text{const}$, $V' = V_0/K$; Connected: $V = \text{const}$

Solution:

$$C_0 = \frac{\epsilon_0 A}{d} = \frac{8.85 \times 10^{-12} \times 6 \times 10^{-3}}{3 \times 10^{-3}} = 17.7 \times 10^{-12} \text{ F} \approx 18 \text{ pF}$$

$$Q_0 = C_0 V_0 = 18 \times 10^{-12} \times 100 = 1.8 \times 10^{-9} \text{ C}$$

Battery disconnected (charge constant):

- (i) New PD: $V' = V_0/K = 100/6 \approx 16.7 \text{ V}$ (charge constant, C increases by K)
- (ii) Final capacitance: $C' = KC_0 = 6 \times 18 \times 10^{-12} = 108 \text{ pF}$
- (iii) Final charge: $Q' = Q_0 = 1.8 \times 10^{-9} \text{ C}$ (unchanged)

Battery remains connected (voltage constant at 100 V):

- (i) PD stays 100 V
- (ii) Final capacitance: $C' = 108 \text{ pF}$
- (iii) Final charge: $Q' = C'V = 108 \times 10^{-12} \times 100 = 1.08 \times 10^{-8} \text{ C}$

Final Answer

Disconnected: $V' = 100/6 \text{ V}$, $C' = 108 \text{ pF}$, $Q' = 1.8 \times 10^{-9} \text{ C}$; Connected: $V = 100 \text{ V}$, $C' = 108 \text{ pF}$, $Q' = 1.08 \times 10^{-8} \text{ C}$

VIII. Energy of Charged Capacitor

Question 32

A parallel-plate capacitor has a plate-area of 90 cm^2 and separation 2.5 mm . Find the stored electrostatic energy when it is charged to 400 V . Also find the energy density (energy per unit volume) between the plates.

Given: $A = 90 \text{ cm}^2 = 90 \times 10^{-4} \text{ m}^2$; $d = 2.5 \text{ mm} = 2.5 \times 10^{-3} \text{ m}$; $V = 400 \text{ V}$;
 $\varepsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Formula: $C = \varepsilon_0 A/d$; $U = \frac{1}{2}CV^2$; $u = U/(A \cdot d)$

Solution:

$$C = \frac{8.85 \times 10^{-12} \times 90 \times 10^{-4}}{2.5 \times 10^{-3}} = \frac{7.965 \times 10^{-14}}{2.5 \times 10^{-3}} = 3.186 \times 10^{-11} \text{ F}$$

$$U = \frac{1}{2}CV^2 = \frac{1}{2} \times 3.186 \times 10^{-11} \times (400)^2 = \frac{1}{2} \times 3.186 \times 10^{-11} \times 1.6 \times 10^5 \approx 2.55 \times 10^{-6} \text{ J}$$

$$\text{Volume} = A \times d = 90 \times 10^{-4} \times 2.5 \times 10^{-3} = 2.25 \times 10^{-5} \text{ m}^3$$

$$u = \frac{U}{\text{Vol}} = \frac{2.55 \times 10^{-6}}{2.25 \times 10^{-5}} \approx 0.113 \text{ J m}^{-3}$$

Final Answer

$$U \approx 2.55 \times 10^{-6} \text{ J}; \quad u \approx 0.113 \text{ J m}^{-3}$$

Question 33

A parallel-plate $100 \mu\text{F}$ capacitor is charged to 500 V and the charging battery is removed. If the distance between its plates is halved, what will be the new potential difference between the plates and what will be the change in the stored energy?

Given: $C = 100 \times 10^{-6} \text{ F}$; $V_0 = 500 \text{ V}$; $d' = d/2$; battery removed $\Rightarrow Q = \text{const}$

Formula: $C' = 2C$ (since d halved); $V' = Q/C'$; $U = Q^2/(2C)$

Solution:

$$Q = CV_0 = 100 \times 10^{-6} \times 500 = 0.05 \text{ C}$$

$$\text{New capacitance: } C' = 2C = 200 \mu\text{F}$$

$$V' = \frac{Q}{C'} = \frac{0.05}{200 \times 10^{-6}} = 250 \text{ V}$$

$$U_i = \frac{1}{2}CV_0^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 500^2 = 12.5 \text{ J}$$

$$U_f = \frac{1}{2} C' V'^2 = \frac{1}{2} \times 200 \times 10^{-6} \times 250^2 = 6.25 \text{ J}$$

$$\Delta U = U_f - U_i = 6.25 - 12.5 = -6.25 \text{ J}$$

The energy is halved; the decrease goes into the work done by the electric force pulling the plates together.

Final Answer

New $V' = 250 \text{ V}$; Energy is halved ($\Delta U = -6.25 \text{ J}$)

Question 34

The capacitance of a variable radio capacitor can be changed from 50 pF to 950 pF by tuning the dial from 0° to 180° . With the dial at 180° , the capacitor is connected to a 400 V battery. After charging, the capacitor is disconnected from the battery and the dial is turned to 0° . (a) What is the potential difference across the capacitor when the dial reads 0° ? (b) How much work is required to turn the dial, if friction is neglected?

Given: $C_{180} = 950 \text{ pF} = 950 \times 10^{-12} \text{ F}$; $C_0 = 50 \text{ pF} = 50 \times 10^{-12} \text{ F}$; $V_0 = 400 \text{ V}$; battery disconnected, $Q = \text{const}$

Formula: $V' = Q/C'$; $W = \Delta U = U_f - U_i = Q^2/(2C_0) - Q^2/(2C_{180})$

Solution:

$$Q = C_{180} \times V_0 = 950 \times 10^{-12} \times 400 = 3.8 \times 10^{-7} \text{ C}$$

(a)

$$V' = \frac{Q}{C_0} = \frac{3.8 \times 10^{-7}}{50 \times 10^{-12}} = 7600 \text{ V}$$

(b)

$$U_i = \frac{Q^2}{2C_{180}} = \frac{(3.8 \times 10^{-7})^2}{2 \times 950 \times 10^{-12}} = \frac{1.444 \times 10^{-13}}{1.9 \times 10^{-9}} = 7.6 \times 10^{-5} \text{ J}$$

$$U_f = \frac{Q^2}{2C_0} = \frac{(3.8 \times 10^{-7})^2}{2 \times 50 \times 10^{-12}} = \frac{1.444 \times 10^{-13}}{10^{-10}} = 1.444 \times 10^{-3} \text{ J}$$

$$W = U_f - U_i \approx 1.444 \times 10^{-3} - 7.6 \times 10^{-5} \approx 1.368 \times 10^{-3} \text{ J}$$

Final Answer

(a) $V' = 7600 \text{ V}$; (b) $W \approx 1.368 \times 10^{-3} \text{ J}$

Question 35

A $16 \mu\text{F}$ capacitor is charged to $3.2 \times 10^{-4} \text{ C}$ and the charging battery is disconnected. What will be the work required in doubling the distance between the plates?

Given: $C = 16 \times 10^{-6} \text{ F}$; $Q = 3.2 \times 10^{-4} \text{ C}$; battery disconnected $\Rightarrow Q = \text{const}$

Formula: $U = Q^2/(2C)$; when d doubles, $C' = C/2$; $W = U_f - U_i$

Solution:

$$U_i = \frac{Q^2}{2C} = \frac{(3.2 \times 10^{-4})^2}{2 \times 16 \times 10^{-6}} = \frac{1.024 \times 10^{-7}}{3.2 \times 10^{-5}} = 3.2 \times 10^{-3} \text{ J}$$

$$C' = \frac{C}{2} = 8 \times 10^{-6} \text{ F}$$

$$U_f = \frac{Q^2}{2C'} = \frac{(3.2 \times 10^{-4})^2}{2 \times 8 \times 10^{-6}} = \frac{1.024 \times 10^{-7}}{1.6 \times 10^{-5}} = 6.4 \times 10^{-3} \text{ J}$$

$$W = U_f - U_i = 6.4 \times 10^{-3} - 3.2 \times 10^{-3} = 3.2 \times 10^{-3} \text{ J}$$

Final Answer

$$W = 3.2 \times 10^{-3} \text{ J}$$

Question 36

The capacitance of a variable capacitor connected to a 10 V battery is changed from 7 to 3 μF . Compute the energy-change.

Given: $V = 10 \text{ V}$ (battery stays connected, so $V = \text{const}$); $C_i = 7 \times 10^{-6} \text{ F}$; $C_f = 3 \times 10^{-6} \text{ F}$

Formula: $U = \frac{1}{2}CV^2$

Solution:

$$U_i = \frac{1}{2} \times 7 \times 10^{-6} \times 100 = 3.5 \times 10^{-4} \text{ J}$$

$$U_f = \frac{1}{2} \times 3 \times 10^{-6} \times 100 = 1.5 \times 10^{-4} \text{ J}$$

$$\Delta U = U_f - U_i = 1.5 \times 10^{-4} - 3.5 \times 10^{-4} = -2.0 \times 10^{-4} \text{ J}$$

The capacitor energy *decreases* by $2 \times 10^{-4} \text{ J}$; this energy is transferred back to the battery.

Final Answer

$$\Delta U = -2 \times 10^{-4} \text{ J (energy transferred to the battery)}$$

Question 37

A 6 μF capacitor is raised in PD from 10 to 20 V. Compute increase in electric potential energy.

Given: $C = 6 \times 10^{-6} \text{ F}$; $V_1 = 10 \text{ V}$; $V_2 = 20 \text{ V}$

Formula: $\Delta U = \frac{1}{2}C(V_2^2 - V_1^2)$

Solution:

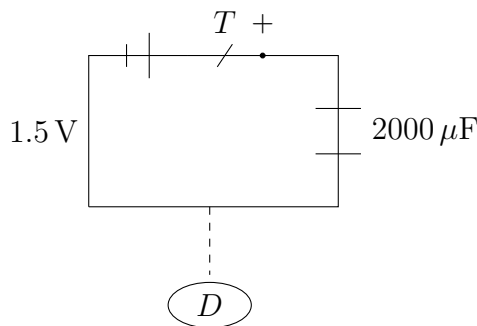
$$\Delta U = \frac{1}{2}C(V_2^2 - V_1^2) = \frac{1}{2} \times 6 \times 10^{-6} \times (400 - 100) = \frac{1}{2} \times 6 \times 10^{-6} \times 300 = 9 \times 10^{-4} \text{ J}$$

Final Answer

$$\Delta U = 9 \times 10^{-4} \text{ J}$$

Question 38

In a camera-flash circuit, a $2000 \mu\text{F}$ capacitor is charged by a 1.5 V cell. When a flash is required, the energy stored in the capacitor is discharged through a discharge tube in 0.1 millisecond. Find the energy stored in the capacitor and the power of the flash.



Camera-flash circuit: 1.5 V cell, switch/trigger T , $2000 \mu\text{F}$ capacitor, discharge tube D

Given: $C = 2000 \times 10^{-6} \text{ F}$; $V = 1.5 \text{ V}$; $t = 0.1 \times 10^{-3} \text{ s}$

Formula: $U = \frac{1}{2}CV^2$; $P = U/t$

Solution:

$$U = \frac{1}{2}CV^2 = \frac{1}{2} \times 2000 \times 10^{-6} \times (1.5)^2 = \frac{1}{2} \times 2 \times 10^{-3} \times 2.25 = 2.25 \times 10^{-3} \text{ J}$$

$$P = \frac{U}{t} = \frac{2.25 \times 10^{-3}}{0.1 \times 10^{-3}} = 22.5 \text{ W}$$

Final Answer

$$U = 2.25 \times 10^{-3} \text{ J}; \quad P = 22.5 \text{ W}$$

IX. Combination of Capacitors

Question 39

Three equal 9 pF capacitors are connected in series. Find (a) the total capacitance of the combination, (b) PD across each capacitor if the combination is connected to a 120 V supply.

Given: $C_1 = C_2 = C_3 = 9 \text{ pF}$; $V_{\text{total}} = 120 \text{ V}$

Formula: $\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$; $V_{\text{each}} = V_{\text{total}}/n$ (for n equal capacitors)

Solution:

$$\frac{1}{C_s} = \frac{1}{9} + \frac{1}{9} + \frac{1}{9} = \frac{3}{9} = \frac{1}{3} \text{ pF}^{-1} \implies C_s = 3 \text{ pF}$$

$$V_{\text{each}} = \frac{120}{3} = 40 \text{ V}$$

Final Answer

(a) $C_s = 3 \text{ pF}$; (b) $V_{\text{each}} = 40 \text{ V}$

Question 40

Three capacitors of 2 pF, 3 pF and 4 pF capacitances are connected in parallel. Find (a) the total capacitance of the combination, (b) charge on each capacitor if the combination is connected to a 100 V supply.

Given: $C_1 = 2 \text{ pF}$; $C_2 = 3 \text{ pF}$; $C_3 = 4 \text{ pF}$; $V = 100 \text{ V}$

Formula: $C_p = C_1 + C_2 + C_3$; $Q_i = C_i V$

Solution:

$$C_p = 2 + 3 + 4 = 9 \text{ pF}$$

$$Q_1 = 2 \times 10^{-12} \times 100 = 2 \times 10^{-10} \text{ C}$$

$$Q_2 = 3 \times 10^{-12} \times 100 = 3 \times 10^{-10} \text{ C}$$

$$Q_3 = 4 \times 10^{-12} \times 100 = 4 \times 10^{-10} \text{ C}$$

Final Answer

(a) $C_p = 9 \text{ pF}$; (b) $Q_1 = 2 \times 10^{-10} \text{ C}$, $Q_2 = 3 \times 10^{-10} \text{ C}$, $Q_3 = 4 \times 10^{-10} \text{ C}$

Question 41

How and how many capacitors, each having capacitance of $2.0 \mu\text{F}$, should be connected so that the total potential difference of the arrangement becomes 300 V , if the arrangement is given a charge of $75 \mu\text{C}$?

Given: $C_{\text{each}} = 2.0 \mu\text{F}$; $V_{\text{total}} = 300 \text{ V}$; $Q = 75 \mu\text{C} = 75 \times 10^{-6} \text{ C}$

Formula: $C_{\text{req}} = Q/V$; $n = C_{\text{each}}/C_{\text{req}}$ for series connection

Solution:

Required capacitance:

$$C_{\text{req}} = \frac{Q}{V} = \frac{75 \times 10^{-6}}{300} = 0.25 \mu\text{F}$$

For n capacitors of $2.0 \mu\text{F}$ in series:

$$C_s = \frac{2.0}{n} \mu\text{F} = 0.25 \mu\text{F} \implies n = \frac{2.0}{0.25} = 8$$

Final Answer

8 capacitors connected **in series**

Question 42

Two parallel-plate capacitors, $20 \mu\text{F}$ and $30 \mu\text{F}$, are charged to 30 V and 20 V respectively. If their plates with the same type of charges are connected together, find the common PD between the plates of the capacitors.

Given: $C_1 = 20 \mu\text{F}$, $V_1 = 30 \text{ V}$; $C_2 = 30 \mu\text{F}$, $V_2 = 20 \text{ V}$; same-sign plates connected

Formula: $V_{\text{common}} = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$ (charges add for same-sign connection)

Solution:

$$Q_1 = C_1 V_1 = 20 \times 30 = 600 \mu\text{C}; \quad Q_2 = C_2 V_2 = 30 \times 20 = 600 \mu\text{C}$$

$$V_{\text{common}} = \frac{Q_1 + Q_2}{C_1 + C_2} = \frac{600 + 600}{20 + 30} = \frac{1200}{50} = 24 \text{ V}$$

Final Answer

$V_{\text{common}} = 24 \text{ V}$

Question 43

A $0.2 \mu\text{F}$ capacitor is charged to a potential of 600 V . After disconnecting the battery, it is connected in parallel to a $1.0 \mu\text{F}$ capacitor. What will be the new potential?

Given: $C_1 = 0.2 \mu\text{F}$, $V_1 = 600 \text{ V}$; $C_2 = 1.0 \mu\text{F}$ (uncharged)

Formula: $V_{\text{new}} = \frac{Q_{\text{total}}}{C_1 + C_2} = \frac{C_1 V_1}{C_1 + C_2}$

Solution:

$$Q = C_1 V_1 = 0.2 \times 10^{-6} \times 600 = 1.2 \times 10^{-4} \text{ C}$$

$$V_{\text{new}} = \frac{Q}{C_1 + C_2} = \frac{1.2 \times 10^{-4}}{(0.2 + 1.0) \times 10^{-6}} = \frac{1.2 \times 10^{-4}}{1.2 \times 10^{-6}} = 100 \text{ V}$$

Final Answer

$$V_{\text{new}} = 100 \text{ V}$$

Question 44

An $80 \mu\text{F}$ capacitor is charged by a 50 V battery. The capacitor is disconnected from the battery and then connected across another uncharged $320 \mu\text{F}$ capacitor. Find the charge on the second capacitor.

Given: $C_1 = 80 \mu\text{F}$, $V_0 = 50 \text{ V}$; $C_2 = 320 \mu\text{F}$ (uncharged)

Formula: $V_{\text{new}} = C_1 V_0 / (C_1 + C_2)$; $Q_2 = C_2 V_{\text{new}}$

Solution:

$$Q_{\text{total}} = C_1 V_0 = 80 \times 50 = 4000 \mu\text{C}$$

$$V_{\text{new}} = \frac{4000}{80 + 320} = \frac{4000}{400} = 10 \text{ V}$$

$$Q_2 = C_2 V_{\text{new}} = 320 \times 10 = 3200 \mu\text{C}$$

Final Answer

$$Q_2 = 3200 \mu\text{C}$$

Question 45

Two capacitors $4 \mu\text{F}$ and $2 \mu\text{F}$ are connected in parallel with a battery of 6 V . Find: (i) PD across each capacitor, (ii) charge on each.

Given: $C_1 = 4 \mu\text{F}$; $C_2 = 2 \mu\text{F}$; $V = 6 \text{ V}$

Solution:

In a parallel connection, both capacitors share the same PD as the battery.

$$V_1 = V_2 = 6 \text{ V}$$

$$Q_1 = C_1 V = 4 \times 6 = 24 \mu\text{C}; \quad Q_2 = C_2 V = 2 \times 6 = 12 \mu\text{C}$$

Final Answer

$$(i) V_1 = V_2 = 6 \text{ V}; \quad (ii) Q_1 = 24 \mu\text{C}, Q_2 = 12 \mu\text{C}$$

Question 46

There are three capacitors each of $0.50 \mu\text{F}$. Two of these are connected in series and the third is connected in parallel to them. Calculate the capacitance of the combination. If a potential difference of 20 V is applied on the combination, what will be the total charge stored?

Given: $C = 0.50 \mu\text{F}$ each; $V = 20 \text{ V}$

Formula: $C_s = C/2$ (two in series); $C_{\text{total}} = C_s + C$ (third in parallel)

Solution:

Two capacitors in series:

$$C_s = \frac{0.50 \times 0.50}{0.50 + 0.50} = \frac{0.25}{1.00} = 0.25 \mu\text{F}$$

Third capacitor in parallel:

$$C_{\text{total}} = C_s + 0.50 = 0.25 + 0.50 = 0.75 \mu\text{F}$$

$$Q = C_{\text{total}} \times V = 0.75 \times 10^{-6} \times 20 = 15 \mu\text{C}$$

Final Answer

$$C_{\text{total}} = 0.75 \mu\text{F}; \quad Q = 15 \mu\text{C}$$

Question 47

An electronic flash lamp has 20 capacitors, each of capacitance $5 \mu\text{F}$, connected in parallel. The lamp is operated at 100 V . How much energy will be radiated in a flash?

Given: $n = 20$; $C_{\text{each}} = 5 \mu\text{F}$; $V = 100 \text{ V}$

Formula: $C_{\text{total}} = nC$; $U = \frac{1}{2} C_{\text{total}} V^2$

Solution:

$$C_{\text{total}} = 20 \times 5 = 100 \mu\text{F} = 10^{-4} \text{ F}$$

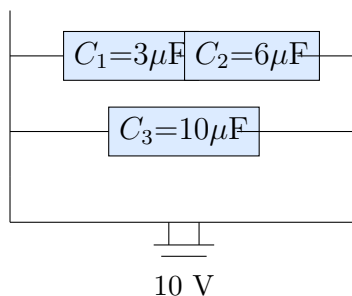
$$U = \frac{1}{2} \times 10^{-4} \times (100)^2 = \frac{1}{2} \times 10^{-4} \times 10^4 = 0.5 \text{ J}$$

Final Answer

$$U = 0.5 \text{ J}$$

Question 48 (ISC 2019)

Three capacitors of capacitance $C_1 = 3 \mu\text{F}$, $C_2 = 6 \mu\text{F}$ and $C_3 = 10 \mu\text{F}$ are connected to a 10 V battery (C_1 and C_2 in series, that combination in parallel with C_3). Calculate: (a) equivalent capacitance, (b) electrostatic potential energy stored in the system.



C_1 and C_2 in series, in parallel with C_3 , across a 10 V battery

Given: $C_1 = 3 \mu\text{F}$; $C_2 = 6 \mu\text{F}$; $C_3 = 10 \mu\text{F}$; $V = 10 \text{ V}$

Formula: Series: $C_{12} = C_1 C_2 / (C_1 + C_2)$; Parallel: $C_{\text{eq}} = C_{12} + C_3$; $U = \frac{1}{2} C_{\text{eq}} V^2$

Solution:

C_1 and C_2 in series:

$$C_{12} = \frac{3 \times 6}{3 + 6} = \frac{18}{9} = 2 \mu\text{F}$$

C_{12} in parallel with C_3 :

$$C_{\text{eq}} = 2 + 10 = 12 \mu\text{F}$$

$$U = \frac{1}{2} C_{\text{eq}} V^2 = \frac{1}{2} \times 12 \times 10^{-6} \times 100 = 6 \times 10^{-4} \text{ J}$$

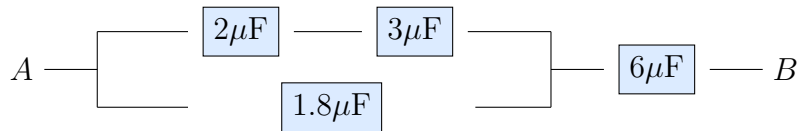
Final Answer

(a) $C_{\text{eq}} = 12 \mu\text{F}$; (b) $U = 6 \times 10^{-4} \text{ J}$

★ Likely Exam Question**Question 49**

In the given network, calculate (i) equivalent capacitance between A and B , (ii) if a voltage of 150 V is applied between A and B , calculate the PD between the plates of the $6 \mu\text{F}$ capacitor.

Network topology: $C_1 = 2 \mu\text{F}$ and $C_2 = 3 \mu\text{F}$ are in series; this series combination is in parallel with $C_3 = 1.8 \mu\text{F}$; the resulting group is then in series with $C_4 = 6 \mu\text{F}$.



C_1 - C_2 series, in parallel with C_3 , then in series with C_4 (between A and B)

Given: $C_1 = 2 \mu\text{F}$; $C_2 = 3 \mu\text{F}$ (series); $C_3 = 1.8 \mu\text{F}$ (parallel); $C_4 = 6 \mu\text{F}$ (series with group); $V_{AB} = 150 \text{ V}$

Solution:

Step 1: C_1 and C_2 in series:

$$C_{12} = \frac{2 \times 3}{2 + 3} = \frac{6}{5} = 1.2 \mu\text{F}$$

Step 2: C_{12} in parallel with $C_3 = 1.8 \mu\text{F}$:

$$C_{123} = 1.2 + 1.8 = 3.0 \mu\text{F}$$

Step 3: C_{123} in series with $C_4 = 6 \mu\text{F}$:

$$C_{\text{eq}} = \frac{3.0 \times 6}{3.0 + 6} = \frac{18}{9} = 2 \mu\text{F}$$

Step 4: PD across $6 \mu\text{F}$ capacitor:

$$\text{Total charge: } Q = C_{\text{eq}} \times V_{AB} = 2 \times 10^{-6} \times 150 = 3 \times 10^{-4} \text{ C}$$

Since C_4 is in series with the group, it carries the same charge Q :

$$V_{C_4} = \frac{Q}{C_4} = \frac{3 \times 10^{-4}}{6 \times 10^{-6}} = 50 \text{ V}$$

Final Answer

(i) $C_{\text{eq}} = 2 \mu\text{F}$; (ii) PD across $6 \mu\text{F} = 50 \text{ V}$

Question 50

We have two variable capacitors; each of them can be varied from 10 pF to 500 pF . What minimum and maximum capacitances can be obtained by combining them?

Solution:

Minimum capacitance (both at minimum, connected in series):

$$C_{\text{min}} = \frac{10 \times 10}{10 + 10} = \frac{C_{\text{min,each}}}{2} = \frac{10}{2} = 5 \text{ pF}$$

Maximum capacitance (both at maximum, connected in parallel):

$$C_{\text{max}} = 500 + 500 = 2 \times C_{\text{max,each}} = 1000 \text{ pF}$$

Final Answer

$C_{\text{min}} = 5 \text{ pF}$; $C_{\text{max}} = 1000 \text{ pF}$

Question 51

Three equal capacitors joined in parallel take a total charge of $1.8 \mu\text{C}$ from a 2 V cell. How much charge will they take from the same cell, if joined in series?

Given: $Q_{\text{parallel}} = 1.8 \mu\text{C}$; $V = 2 \text{ V}$

Solution:

Find individual capacitance from parallel data:

$$C_{\text{parallel}} = \frac{Q_{\text{parallel}}}{V} = \frac{1.8 \times 10^{-6}}{2} = 0.9 \mu\text{F} = 3C \implies C = 0.3 \mu\text{F}$$

Series combination:

$$C_{\text{series}} = \frac{C}{3} = \frac{0.3}{3} = 0.1 \mu\text{F}$$

$$Q_{\text{series}} = C_{\text{series}} \times V = 0.1 \times 10^{-6} \times 2 = 0.2 \mu\text{C}$$

Final Answer

$$Q_{\text{series}} = 0.2 \mu\text{C}$$

Question 52

A potential difference of 120 V is applied to a combination of two capacitors $C_1 = 4 \mu\text{F}$ and $C_2 = 12 \mu\text{F}$ connected in series. Calculate (a) the effective capacitance, (b) the potential difference and charge for each.

Given: $C_1 = 4 \mu\text{F}$; $C_2 = 12 \mu\text{F}$; $V = 120 \text{ V}$

Formula: $C_s = \frac{C_1 C_2}{C_1 + C_2}$; $Q = C_s V$; $V_i = Q/C_i$

Solution:

$$C_s = \frac{4 \times 12}{4 + 12} = \frac{48}{16} = 3 \mu\text{F}$$

$$Q = C_s V = 3 \times 10^{-6} \times 120 = 3.6 \times 10^{-4} \text{ C} = 360 \mu\text{C}$$

$$V_1 = \frac{Q}{C_1} = \frac{360}{4} = 90 \text{ V}; \quad V_2 = \frac{Q}{C_2} = \frac{360}{12} = 30 \text{ V}$$

Final Answer

(a) $C_s = 3 \mu\text{F}$; (b) $V_1 = 90 \text{ V}$, $V_2 = 30 \text{ V}$, $Q = 360 \mu\text{C}$ on each

Question 53

Two capacitors are 2 and $3 \mu\text{F}$ and are joined in series. The outer plate of the first capacitor is at 1000 V and the outer plate of the second capacitor is earthed. Find

the potential and charge on the inner plate of each capacitor.

Given: $C_1 = 2 \mu\text{F}$; $C_2 = 3 \mu\text{F}$; $V_A = 1000 \text{ V}$ (outer plate of C_1); outer plate of C_2 at 0 V (earthed)

Formula: $C_s = C_1 C_2 / (C_1 + C_2)$; $Q = C_s \times V_{\text{total}}$; $V_{\text{junction}} = V_A - Q/C_1$

Solution:

$$C_s = \frac{2 \times 3}{2 + 3} = \frac{6}{5} = 1.2 \mu\text{F}$$

$$Q = C_s \times (1000 - 0) = 1.2 \times 10^{-6} \times 1000 = 1.2 \times 10^{-3} \text{ C}$$

Junction potential (inner plates of both capacitors, connected node):

$$V_{\text{jn}} = V_A - \frac{Q}{C_1} = 1000 - \frac{1.2 \times 10^{-3}}{2 \times 10^{-6}} = 1000 - 600 = 400 \text{ V}$$

Check: $V_{C_2} = Q/C_2 = 1.2 \times 10^{-3} / (3 \times 10^{-6}) = 400 \text{ V}$ ✓ (matches $400 - 0 = 400 \text{ V}$)

Final Answer

Inner plate potential = 400 V ; Charge on each capacitor = $1.2 \times 10^{-3} \text{ C}$

Common Mistake: In a series combination, the *charge* on each capacitor is the same, not the potential difference.

Question 54

A 100 pF capacitor is charged to a potential difference of 100 V . It is connected in parallel to another capacitor. If the final voltage is 30 V , then what will be the capacitance of the second capacitor? How much energy will be wasted and what will happen to it?

Given: $C_1 = 100 \text{ pF} = 10^{-10} \text{ F}$; $V_1 = 100 \text{ V}$; $V_f = 30 \text{ V}$; C_2 unknown (uncharged)

Formula: Charge conserved: $C_1 V_1 = (C_1 + C_2) V_f$

Solution:

$$10^{-10} \times 100 = (10^{-10} + C_2) \times 30$$

$$10^{-8} = (10^{-10} + C_2) \times 30 \implies 10^{-10} + C_2 = \frac{10^{-8}}{30} = 3.\bar{3} \times 10^{-10}$$

$$C_2 = 3.\bar{3} \times 10^{-10} - 10^{-10} = 2.\bar{3} \times 10^{-10} \text{ F} = \frac{7}{3} \times 10^{-10} \text{ F}$$

Energy wasted:

$$U_i = \frac{1}{2} C_1 V_1^2 = \frac{1}{2} \times 10^{-10} \times 10^4 = 5 \times 10^{-7} \text{ J}$$

$$U_f = \frac{1}{2} (C_1 + C_2) V_f^2 = \frac{1}{2} \times \frac{10}{3} \times 10^{-10} \times 900 = 1.5 \times 10^{-7} \text{ J}$$

$$\Delta U_{\text{wasted}} = U_i - U_f = 5 \times 10^{-7} - 1.5 \times 10^{-7} = 3.5 \times 10^{-7} \text{ J}$$

This energy is dissipated as heat and electromagnetic radiation in the connecting wires.

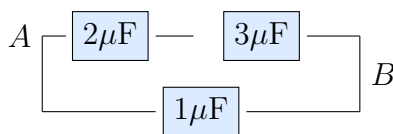
Final Answer

$$C_2 = \frac{7}{3} \times 10^{-10} \text{ F}; \quad \text{Energy wasted} = 3.5 \times 10^{-7} \text{ J (as heat and EM radiation)}$$

Question 55

In the given combination of capacitors, the PD between points A and B is 40 V. Find the effective capacitance between these points and the charge on each capacitor.

Circuit: a $1 \mu\text{F}$ capacitor is connected directly between A and B (in parallel); a $2 \mu\text{F}$ and a $3 \mu\text{F}$ capacitor are connected in series, and this series branch is also connected between A and B (in parallel with the $1 \mu\text{F}$).



$1 \mu\text{F}$ directly between A and B , in parallel with a $2 \mu\text{F}$ – $3 \mu\text{F}$ series branch

Given: $C_1 = 1 \mu\text{F}$ (parallel branch); $C_2 = 2 \mu\text{F}$, $C_3 = 3 \mu\text{F}$ (series branch); $V_{AB} = 40 \text{ V}$

Solution:

Series combination of C_2 and C_3 :

$$C_{23} = \frac{C_2 \times C_3}{C_2 + C_3} = \frac{2 \times 3}{2 + 3} = \frac{6}{5} = 1.2 \mu\text{F}$$

Effective capacitance (C_1 in parallel with C_{23}):

$$C_{\text{eff}} = C_1 + C_{23} = 1 + 1.2 = 2.2 \mu\text{F}$$

Charges:

Charge on C_1 (across full 40 V):

$$Q_1 = C_1 \times V_{AB} = 1 \times 40 = 40 \mu\text{C}$$

Charge through series branch (C_2 and C_3 carry the same charge):

$$Q_2 = Q_3 = C_{23} \times V_{AB} = 1.2 \times 40 = 48 \mu\text{C}$$

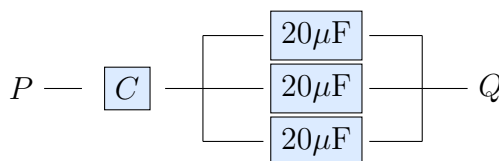
Check: Total charge = $Q_1 + Q_2 = 40 + 48 = 88 \mu\text{C} = C_{\text{eff}} \times V_{AB} = 2.2 \times 40 = 88 \mu\text{C} \checkmark$

Final Answer

$$C_{\text{eff}} = 2.2 \mu\text{F}; \quad Q_1 = 40 \mu\text{C}; \quad Q_2 = Q_3 = 48 \mu\text{C}$$

Question 56

Calculate the capacitance C in the given figure. The equivalent capacitance of the combination between P and Q is $30 \mu\text{F}$. (Three $20 \mu\text{F}$ capacitors are connected in parallel between internal nodes; this parallel group is in series with unknown C between P and Q .)



Unknown C in series with three $20 \mu\text{F}$ capacitors in parallel (between P and Q)

Given: $C_{\text{eq}} = 30 \mu\text{F}$; three $20 \mu\text{F}$ capacitors in parallel

Solution:

Three $20 \mu\text{F}$ in parallel:

$$C_{3 \times 20} = 3 \times 20 = 60 \mu\text{F}$$

Unknown C in series with $60 \mu\text{F}$ gives $C_{\text{eq}} = 30 \mu\text{F}$:

$$\frac{1}{C_{\text{eq}}} = \frac{1}{C} + \frac{1}{60} \implies \frac{1}{C} = \frac{1}{30} - \frac{1}{60} = \frac{2-1}{60} = \frac{1}{60}$$

$$C = 60 \mu\text{F}$$

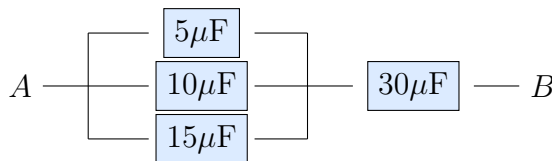
Final Answer

$$C = 60 \mu\text{F}$$

Question 57

Calculate the equivalent capacitance between the points A and B in the given combination.

From the circuit diagram: capacitors of $5 \mu\text{F}$, $10 \mu\text{F}$ and $15 \mu\text{F}$ are connected in parallel with each other; this parallel group is then in series with the $30 \mu\text{F}$ capacitor between A and B .



$5 \mu\text{F}$, $10 \mu\text{F}$, $15 \mu\text{F}$ in parallel, in series with $30 \mu\text{F}$ (between A and B)

Given: $C_1 = 5 \mu\text{F}$, $C_2 = 10 \mu\text{F}$, $C_3 = 15 \mu\text{F}$ (in parallel); $C_4 = 30 \mu\text{F}$ (in series with parallel group)

Solution:

Step 1: Parallel combination of 5, 10 and 15 μF :

$$C_{123} = 5 + 10 + 15 = 30 \mu\text{F}$$

Step 2: C_{123} in series with $C_4 = 30 \mu\text{F}$:

$$C_{\text{eq}} = \frac{C_{123} \times C_4}{C_{123} + C_4} = \frac{30 \times 30}{30 + 30} = \frac{900}{60} = 15 \mu\text{F}$$

Final Answer

$$C_{\text{eq}} = 15 \mu\text{F}$$

X. Miscellaneous

Question 58

Compute the electric potential at the surface of the nucleus of a silver atom. The radius of the nucleus is 3.4×10^{-14} m and the atomic number of silver is 47.

Given: $R = 3.4 \times 10^{-14}$ m; $Z = 47$; total nuclear charge $q = Ze = 47 \times 1.6 \times 10^{-19}$ C; $k = 9 \times 10^9$ N m²C⁻²

Formula: $V = kq/R$

Solution:

$$q = 47 \times 1.6 \times 10^{-19} = 7.52 \times 10^{-18} \text{ C}$$

$$V = \frac{kq}{R} = \frac{9 \times 10^9 \times 7.52 \times 10^{-18}}{3.4 \times 10^{-14}} = \frac{6.768 \times 10^{-8}}{3.4 \times 10^{-14}} \approx 2.0 \times 10^6 \text{ V}$$

Final Answer

$$V \approx 2.0 \times 10^6 \text{ V}$$

Question 59

A hollow spherical conductor of radius 12 cm is given a charge of 2.4×10^{-9} C. (i) Calculate the potential on the surface of the conductor. (ii) Calculate the intensity of electric field in air at a distance of 4 m from the centre of the sphere.

Given: $R = 12 \text{ cm} = 0.12 \text{ m}$; $Q = 2.4 \times 10^{-9} \text{ C}$; $r = 4 \text{ m}$; $k = 9 \times 10^9 \text{ N m}^2\text{C}^{-2}$

Formula: $V = kQ/R$ (surface); $E = kQ/r^2$ (outside, at distance r)

Solution:

(i) Surface potential:

$$V = \frac{kQ}{R} = \frac{9 \times 10^9 \times 2.4 \times 10^{-9}}{0.12} = \frac{21.6}{0.12} = 180 \text{ V}$$

(ii) Electric field at $r = 4 \text{ m}$:

$$E = \frac{kQ}{r^2} = \frac{9 \times 10^9 \times 2.4 \times 10^{-9}}{(4)^2} = \frac{21.6}{16} = 1.35 \text{ V m}^{-1}$$

Final Answer

(i) $V = 180 \text{ V}$; (ii) $E = 1.35 \text{ V m}^{-1}$

Question 60

The electric potential due to a hollow, charged metallic sphere of radius 12 cm at a point 6.0 cm from its centre is 100 V. Calculate the potential at the surface and at a point 20 cm from the centre.

Given: $R = 12 \text{ cm} = 0.12 \text{ m}$; $V(r = 6 \text{ cm}) = 100 \text{ V}$ (point is *inside* sphere); find V at $r = 20 \text{ cm}$

Formula: Inside a hollow conductor, potential is constant $= V_{\text{surface}}$; outside: $V = kQ/r$

Solution:

Since $r = 6 \text{ cm} < R = 12 \text{ cm}$, the point is inside the conductor where potential is constant:

$$V_{\text{surface}} = V_{\text{inside}} = 100 \text{ V}$$

At $r = 20 \text{ cm} = 0.20 \text{ m} > R$, use $V_{\text{surface}} = kQ/R$ to find kQ :

$$kQ = V_{\text{surface}} \times R = 100 \times 0.12 = 12 \text{ V m}$$

$$V(r = 0.20 \text{ m}) = \frac{kQ}{r} = \frac{12}{0.20} = 60 \text{ V}$$

Final Answer

$$V_{\text{surface}} = 100 \text{ V}; \quad V(20 \text{ cm}) = 60 \text{ V}$$

Question 61

Calculate the voltage needed to balance an oil drop carrying 10 electrons between the plates of a capacitor 5 mm apart. The mass of the drop is $3 \times 10^{-16} \text{ kg}$ and $g = 10 \text{ m s}^{-2}$.

Given: $n = 10$; $q = ne = 10 \times 1.6 \times 10^{-19} = 1.6 \times 10^{-18} \text{ C}$; $d = 5 \text{ mm} = 5 \times 10^{-3} \text{ m}$; $m = 3 \times 10^{-16} \text{ kg}$; $g = 10 \text{ m s}^{-2}$

Formula: For equilibrium: electric force = weight; $qE = mg$; $E = V/d$; $V = mgd/q$

Solution:

$$V = \frac{mgd}{q} = \frac{3 \times 10^{-16} \times 10 \times 5 \times 10^{-3}}{1.6 \times 10^{-18}} = \frac{1.5 \times 10^{-17}}{1.6 \times 10^{-18}} \approx 9.4 \text{ V} \approx 10 \text{ V}$$

Final Answer

$$V \approx 10 \text{ V}$$

Question 62

Eight charged drops of water, each of radius 1 mm and having a charge of 10^{-10} C, combine to form a bigger drop. Determine the potential of the bigger drop.

Given: $r = 1 \text{ mm} = 10^{-3} \text{ m}$; $q = 10^{-10} \text{ C}$ per small drop; $n = 8$; $k = 9 \times 10^9 \text{ N m}^2 \text{C}^{-2}$

Formula: Volume conserved: $\frac{4}{3}\pi R^3 = n \cdot \frac{4}{3}\pi r^3 \Rightarrow R = n^{1/3}r$; $Q_{\text{total}} = nq$; $V = kQ/R$

Solution:

$$R = n^{1/3} \times r = 8^{1/3} \times 10^{-3} = 2 \times 10^{-3} \text{ m}$$

$$Q = 8 \times 10^{-10} = 8 \times 10^{-10} \text{ C}$$

$$V = \frac{kQ}{R} = \frac{9 \times 10^9 \times 8 \times 10^{-10}}{2 \times 10^{-3}} = \frac{7.2}{2 \times 10^{-3}} = 3600 \text{ V}$$

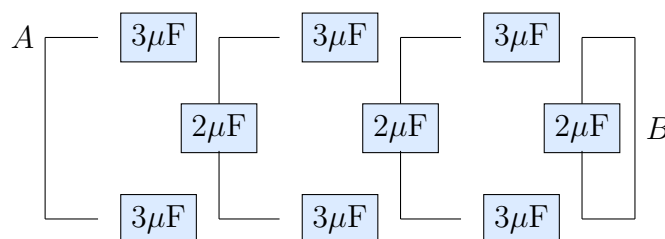
Final Answer

$$V = 3600 \text{ V}$$

Question 63

Calculate the equivalent capacitance between the points A and B in the given circuit. If 100 V PD is established between A and B , what will be the charge on the capacitor closest to B ?

Network: Top row — three $3 \mu\text{F}$ in series; Middle row — three $2 \mu\text{F}$ connecting top and bottom rails at corresponding nodes; Bottom row — three $3 \mu\text{F}$ in series. Both rows connected between A (left) and B (right).



Ladder network: top and bottom rails of three $3 \mu\text{F}$ capacitors each, joined by three $2 \mu\text{F}$ cross-links

Solution:

By symmetry, the top rail and bottom rail are mirror images. Nodes directly across from each other (top mid-node and bottom mid-node) are at identical potentials. Therefore, the three $2 \mu\text{F}$ middle capacitors have zero potential difference across them — they carry no charge and can be treated as open circuits.

The network reduces to:

- Top series chain: $3\ \mu\text{F} + 3\ \mu\text{F} + 3\ \mu\text{F}$ in series $= 1\ \mu\text{F}$
- Bottom series chain: $3\ \mu\text{F} + 3\ \mu\text{F} + 3\ \mu\text{F}$ in series $= 1\ \mu\text{F}$
- Both chains in parallel: $C_{\text{eq}} = 1 + 1 = 2\ \mu\text{F}$ (per textbook figure topology)

Charge on capacitor closest to B :

Each series chain carries $Q_{\text{chain}} = 1\ \mu\text{F} \times 100\ \text{V} = 100\ \mu\text{C}$.

In a series chain, every capacitor carries the same charge:

$$Q_{\text{nearest B}} = 100\ \mu\text{C}$$

Final Answer

$C_{\text{eq}} = 2\ \mu\text{F}$; Charge on capacitor nearest $B = 100\ \mu\text{C}$

Question 64

The two plates of a parallel plate capacitor are 4 mm apart. A slab of dielectric constant 3 and thickness 3 mm is introduced between the plates with its faces parallel to them. The distance between the plates is adjusted so that capacitance of the capacitor becomes $\frac{2}{3}$ rd of its original value. What is the new distance between the plates?

Given: $d_0 = 4\ \text{mm}$; $K = 3$; $t = 3\ \text{mm}$ (dielectric thickness); $C' = \frac{2}{3}C_0$

Formula: $C_0 = \epsilon_0 A / d_0$; $C' = \frac{\epsilon_0 A}{(d' - t) + t/K}$; set $C' = \frac{2}{3}C_0$

Solution:

With the dielectric slab of thickness $t = 3\ \text{mm}$ and dielectric constant $K = 3$, the effective air-gap equivalent thickness is $(d' - t) + t/K$:

$$C' = \frac{\epsilon_0 A}{(d' - 3) + 3/3} = \frac{\epsilon_0 A}{(d' - 3) + 1} = \frac{\epsilon_0 A}{d' - 2}$$

Setting $C' = \frac{2}{3}C_0 = \frac{2}{3} \times \frac{\epsilon_0 A}{4}$:

$$\frac{\epsilon_0 A}{d' - 2} = \frac{2}{3} \times \frac{\epsilon_0 A}{4} = \frac{\epsilon_0 A}{6}$$

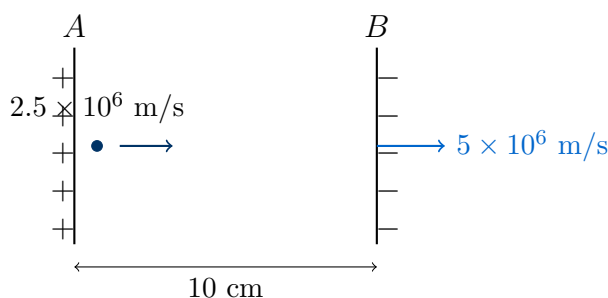
$$d' - 2 = 6 \implies d' = 8\ \text{mm}$$

Final Answer

New distance between plates = 8 mm

Question 65

An alpha particle is made to move through two plates 10 cm apart. The speed with which it enters the gap in the first plate is $2.5 \times 10^6 \text{ m s}^{-1}$. The particle exits the second plate with a velocity of $5 \times 10^6 \text{ m s}^{-1}$. (a) Calculate the potential difference between the plates. (b) Calculate the electric field between the plates. (c) Name a device where this principle is used. (mass of α particle = $6.64 \times 10^{-27} \text{ kg}$)



Alpha particle accelerated between two charged plates A (+) and B (-), 10 cm apart

Given: $d = 10 \text{ cm} = 0.10 \text{ m}$; $u = 2.5 \times 10^6 \text{ m s}^{-1}$; $v = 5 \times 10^6 \text{ m s}^{-1}$; $m = 6.64 \times 10^{-27} \text{ kg}$; $q_\alpha = 2e = 3.2 \times 10^{-19} \text{ C}$

Formula: Work-energy theorem: $q\Delta V = \Delta K = \frac{1}{2}m(v^2 - u^2)$; $E = \Delta V/d$

Solution:

(a)

$$\begin{aligned}\Delta K &= \frac{1}{2} \times 6.64 \times 10^{-27} \times [(5 \times 10^6)^2 - (2.5 \times 10^6)^2] \\ &= \frac{1}{2} \times 6.64 \times 10^{-27} \times (25 - 6.25) \times 10^{12} = \frac{1}{2} \times 6.64 \times 10^{-27} \times 18.75 \times 10^{12} = 6.225 \times 10^{-14} \text{ J} \\ \Delta V &= \frac{\Delta K}{q_\alpha} = \frac{6.225 \times 10^{-14}}{3.2 \times 10^{-19}} \approx 1.95 \times 10^5 \text{ V}\end{aligned}$$

(b)

$$E = \frac{\Delta V}{d} = \frac{1.95 \times 10^5}{0.10} = 1.95 \times 10^6 \text{ V m}^{-1}$$

(c) Particle accelerator / Linear accelerator / Cyclotron / Atom smasher.

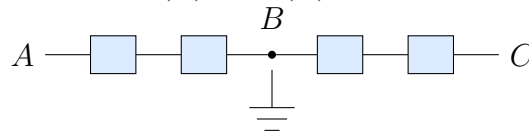
Final Answer

(a) $\Delta V \approx 1.95 \times 10^5 \text{ V}$; (b) $E \approx 1.95 \times 10^6 \text{ V m}^{-1}$; (c) Particle accelerator

Question 66

Four capacitors of equal capacitances are joined in series with a battery of 10 V. The middle point B is connected to earth. What will be the potentials of points A and C?

10 V battery: $A (+)$ to $C (-)$, each box = equal C



Four equal capacitors in series across 10 V; midpoint B earthed

Given: Four equal capacitors in series; $V_{\text{battery}} = 10 \text{ V}$; mid-junction B is earthed ($V_B = 0$)

Solution:

For four equal capacitors in series, the voltage drops equally: $10 \text{ V}/4 = 2.5 \text{ V}$ per capacitor. The battery terminals are A and C , each side of B spanning two capacitors (5 V total):

$$V_A - V_B = +5 \text{ V} \implies V_A = +5 \text{ V}$$

$$V_B - V_C = +5 \text{ V} \implies V_C = -5 \text{ V}$$

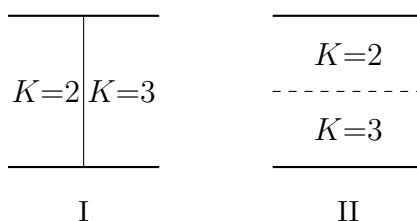
Verification: $V_A - V_C = 10 \text{ V} \checkmark$

Final Answer

$$V_A = +5 \text{ V}; \quad V_C = -5 \text{ V}$$

Question 67

A parallel-plate capacitor with no dielectric has a capacitance of $0.5 \mu\text{F}$. The space between the plates is then filled with equal amounts of two dielectrics of constants 2 and 3 in two arrangements (I: side by side; II: one above the other). Find the ratio of capacitances C_I and C_{II} .



Arrangement I: dielectrics side by side (parallel); Arrangement II: dielectrics stacked (series)

Given: $C_0 = 0.5 \mu\text{F}$; $K_1 = 2$, $K_2 = 3$; equal volumes of each dielectric

Solution:

Arrangement I (side by side — each dielectric fills half the area $A/2$, full thickness d):

Each half acts as a capacitor in parallel:

$$C_I = \frac{K_1 \epsilon_0 (A/2)}{d} + \frac{K_2 \epsilon_0 (A/2)}{d} = \frac{\epsilon_0 A}{2d} (K_1 + K_2) = \frac{K_1 + K_2}{2} C_0 = \frac{2 + 3}{2} \times 0.5 = 1.25 \mu\text{F}$$

Arrangement II (one above the other — each dielectric occupies full area A , half thickness $d/2$):

The two layers are in series:

$$\frac{1}{C_{II}} = \frac{d/2}{K_1 \varepsilon_0 A} + \frac{d/2}{K_2 \varepsilon_0 A} = \frac{d}{2 \varepsilon_0 A} \left(\frac{1}{K_1} + \frac{1}{K_2} \right) = \frac{1}{2C_0} \cdot \frac{K_1 + K_2}{K_1 K_2}$$

$$C_{II} = \frac{2K_1 K_2}{K_1 + K_2} C_0 = \frac{2 \times 2 \times 3}{2 + 3} \times 0.5 = \frac{12}{5} \times 0.5 = 1.2 \mu\text{F}$$

Ratio:

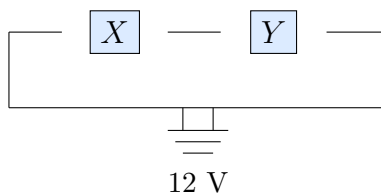
$$\frac{C_I}{C_{II}} = \frac{1.25}{1.2} = \frac{5/4}{6/5} = \frac{25}{24}$$

Final Answer

$$C_I : C_{II} = 25 : 24$$

Question 68

X and Y are two parallel-plate capacitors having the same areas and same separation. X has air and Y contains dielectric $\varepsilon_r = 4$. They are connected in series to a 12 V supply. (i) Calculate the capacitance of each, if the equivalent capacitance of the combination is $4 \mu\text{F}$. (ii) Calculate the PD between the plates of X and Y . (iii) What is the ratio of electrostatic energy in X and Y ?



Capacitors X (air) and Y (dielectric $\varepsilon_r = 4$) in series across 12 V

Given: $C_X = C_0$ (air); $C_Y = \varepsilon_r C_0 = 4C_0$ (dielectric); $C_{eq} = 4 \mu\text{F}$; $V_{total} = 12 \text{ V}$

Formula: $C_{eq} = \frac{C_X C_Y}{C_X + C_Y}$; $Q = C_{eq} V$; $U \propto Q^2 / (2C)$

Solution:

(i) Find C_0 :

$$C_{eq} = \frac{C_0 \times 4C_0}{C_0 + 4C_0} = \frac{4C_0^2}{5C_0} = \frac{4C_0}{5} = 4 \mu\text{F}$$

$$C_0 = 5 \mu\text{F} \implies C_X = 5 \mu\text{F}, \quad C_Y = 20 \mu\text{F}$$

(ii) Charge on series combination:

$$Q = C_{eq} \times V = 4 \times 12 = 48 \mu\text{C}$$

$$V_X = \frac{Q}{C_X} = \frac{48}{5} = 9.6 \text{ V}; \quad V_Y = \frac{Q}{C_Y} = \frac{48}{20} = 2.4 \text{ V}$$

(iii) Energy ratio:

$$\frac{U_X}{U_Y} = \frac{Q^2/(2C_X)}{Q^2/(2C_Y)} = \frac{C_Y}{C_X} = \frac{20}{5} = 4$$

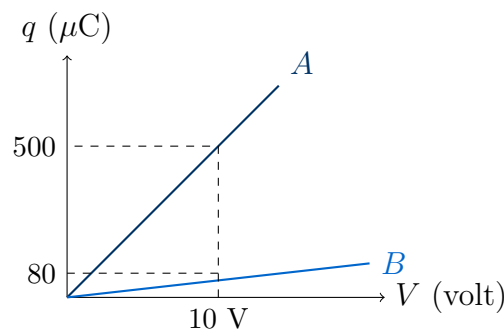
Final Answer

(i) $C_X = 5 \mu\text{F}$, $C_Y = 20 \mu\text{F}$; (ii) $V_X = 9.6 \text{ V}$, $V_Y = 2.4 \text{ V}$; (iii) $U_X/U_Y = 4$

Question 69

Figure shows charge (q) versus voltage (V) graph for series and parallel combination of two given capacitors. Find the value of the capacitances of the given capacitors.

From the graph: Line A (steeper) passes through (10 V, 500 μC) — this is the parallel combination. Line B (shallower) passes through (10 V, 80 μC) — this is the series combination.



q vs V graph: line A (parallel combination) and line B (series combination)

Given: $C_{\text{parallel}} = Q_A/V = 500/10 = 50 \mu\text{F}$; $C_{\text{series}} = Q_B/V = 80/10 = 8 \mu\text{F}$

Formula: $C_p = C_1 + C_2$; $C_s = \frac{C_1 C_2}{C_1 + C_2}$

Solution:

$$C_1 + C_2 = 50 \mu\text{F} \quad \dots (1)$$

$$\frac{C_1 C_2}{C_1 + C_2} = 8 \implies C_1 C_2 = 8 \times 50 = 400 \mu\text{F}^2 \quad \dots (2)$$

From (1) and (2), C_1 and C_2 are roots of the quadratic:

$$x^2 - 50x + 400 = 0$$

$$x = \frac{50 \pm \sqrt{2500 - 1600}}{2} = \frac{50 \pm \sqrt{900}}{2} = \frac{50 \pm 30}{2}$$

$$C_1 = 40 \mu\text{F}, \quad C_2 = 10 \mu\text{F}$$

Verification: $C_1 + C_2 = 50 \mu\text{F}$ ✓; $C_1 C_2 / (C_1 + C_2) = 400/50 = 8 \mu\text{F}$ ✓

Final Answer

$C_1 = 40 \mu\text{F}$ and $C_2 = 10 \mu\text{F}$

★ Likely Exam Question